PCA — Assignment MCQs

1. What is the primary purpose of Principal Component Analysis (PCA)?
    - A. Classification
    - B. Dimensionality reduction
    - C. Data labeling
    - D. Data generation
2. PCA transforms the original variables into:
    - A. Random variables
    - B. Principal components
    - C. Class labels
    - D. Probability distributions
3. Principal components are:
    - A. Always identical to the original features
    - B. Linear combinations of the original features
    - C. Class labels generated from the data
    - D. Randomly selected features
4. What does the first principal component represent?
    - A. The direction of minimum variance
    - B. The direction of maximum variance
    - C. The first feature in the dataset
    - D. The class with the highest frequency
5. What does the second principal component represent?
    - A. The same direction as the first component
    - B. A direction orthogonal to the first component with the next highest variance
    - C. The feature with the second-highest mean
    - D. The second original feature
6. PCA is primarily:
    - A. A supervised learning technique
    - B. An unsupervised learning technique
    - C. A reinforcement learning technique
    - D. A classification algorithm

7. Which type of information does PCA primarily try to preserve?
    - A. Class labels
    - B. Variance
    - C. Missing values
    - D. Feature names
8. Why is PCA useful for high-dimensional datasets?
    - A. It increases the number of features
    - B. It reduces the number of dimensions while retaining important variation
    - C. It automatically labels the data
    - D. It removes all noise
9. In PCA, why is variance important?
    - A. It determines the number of classes
    - B. It indicates the amount of variation captured along a direction
    - C. It determines the data type
    - D. It represents the number of samples
10. Consider a 2D dataset forming an elongated cloud. The first principal component is most likely aligned with:

- A. The shortest direction of the cloud
- B. The longest direction of the cloud
- C. The x-axis always
- D. The y-axis always

11. If all observations lie exactly along a straight line in 2D space, how many principal components are required to represent the data without loss?

- A. 0
- B. 1
- C. 2
- D. 3

12. PCA finds directions that:

- A. Maximize variance successively
- B. Minimize the number of samples
- C. Maximize the number of classes
- D. Minimize the number of features independently

13. What is the geometric interpretation of PCA?

- A. Rotating the coordinate system to find directions of maximum variance
- B. Sorting the samples according to class
- C. Removing all negative values
- D. Increasing the dimensionality of the data

14. If a dataset has very little variation along one direction, PCA is likely to:

- A. Assign that direction a large principal component variance
- B. Assign that direction a small principal component variance
- C. Always select it as PC1
- D. Duplicate that direction

15. Why is mean centering commonly performed before PCA?

- A. To move the data so that its mean is at the origin
- B. To increase the number of features
- C. To convert categorical variables into labels
- D. To remove all correlations

16. What happens if PCA is applied to data without centering when centering is appropriate?

- A. The principal directions may be influenced by the location of the data
- B. PCA becomes supervised
- C. The number of samples changes
- D. All eigenvalues become zero

17. Why might standardization be important before PCA?

- A. To ensure features with larger scales do not dominate the variance
- B. To increase the number of observations
- C. To remove the mean from every sample independently
- D. To create new labels

18. Suppose one feature ranges from 0–10 and another from 0–100,000. Without appropriate scaling, PCA may:

- A. Give excessive importance to the larger-scale feature
- B. Ignore both features
- C. Automatically normalize both features
- D. Remove both features

19. Mean centering changes:

- A. The relative distances between all observations
- B. The location of the data relative to the origin
- C. The number of observations
- D. The number of features

20. PCA is closely related to the analysis of:

- A. Covariance
- B. Class labels
- C. Decision trees
- D. Confusion matrices

21. What does a covariance matrix describe?

- A. Relationships between variables
- B. Class prediction accuracy
- C. Number of training epochs
- D. Number of missing values

22. A covariance matrix for a dataset with 5 features has dimensions:

- A. 5×1
- B. 1×5
- C. 5×5
- D. 10×10

23. The diagonal elements of a covariance matrix represent:

- A. Covariance between different features
- B. Variance of individual features
- C. Class probabilities
- D. Principal component scores

24. If two features have covariance close to zero, this indicates:

- A. They have no linear covariance relationship
- B. They are always identical
- C. They must be independent
- D. They must have equal variance

25. A strong correlation between two features suggests that:

- A. There may be redundant information
- B. They must belong to different classes
- C. PCA cannot be applied
- D. Both features must be removed

26. PCA commonly uses eigenvectors of the covariance matrix to determine:

- A. Principal directions
- B. Class labels
- C. Missing values
- D. Sample sizes

27. In PCA, eigenvalues indicate:

- A. The amount of variance associated with principal components
- B. The number of samples
- C. The number of features
- D. The classification accuracy

28. If PC1 has a much larger eigenvalue than PC2, this suggests:

- A. PC1 captures more variance than PC2
- B. PC2 captures more variance than PC1
- C. Both capture exactly the same variance
- D. PC1 is a class label

29. Which principal component generally corresponds to the largest eigenvalue?

- A. PC1
- B. PC2
- C. PC3
- D. Last principal component

30. Principal components are usually ordered according to:

- A. Increasing sample size
- B. Decreasing explained variance
- C. Alphabetical order of features
- D. Increasing feature index

31. If a covariance matrix has eigenvalues 10, 3, and 1, which component captures the greatest variance?

- A. PC1
- B. PC2
- C. PC3
- D. All equally

32. The eigenvectors corresponding to different eigenvalues of a symmetric covariance matrix are:

- A. Orthogonal
- B. Identical
- C. Random
- D. Dependent on class labels

33. What does explained variance ratio represent?

- A. The proportion of total variance captured by a principal component
- B. The proportion of correctly classified samples
- C. The percentage of missing values
- D. The number of original features

34. If PC1 explains 70% of the variance and PC2 explains 20%, together they explain:

- A. 50%
- B. 70%
- C. 90%
- D. 100%

35. If the first three principal components explain 95% of the variance, this means:

- A. 95% of the original samples are retained
- B. The three components capture 95% of the variance
- C. 95% of the original features are removed
- D. 95% classification accuracy is achieved

36. A scree plot is commonly used to:

- A. Select the number of principal components
- B. Evaluate classification accuracy
- C. Detect missing values
- D. Calculate class labels

37. In a scree plot, an "elbow" generally indicates:

- A. A point beyond which additional components provide relatively smaller gains
- B. The number of classes
- C. The number of samples
- D. The feature with the largest mean

38. A dataset contains 100 features. PCA reduces it to 10 principal components. What has happened?

- A. The dimensionality has been reduced
- B. The number of observations has been reduced
- C. The dataset has become supervised
- D. Ten samples remain

39. What is a major advantage of PCA-based dimensionality reduction?

- A. Lower computational and storage requirements
- B. Guaranteed higher classification accuracy
- C. Guaranteed removal of all noise
- D. Preservation of every original feature independently

40. When PCA reduces dimensions, the resulting components:

- A. Are generally combinations of the original features
- B. Are always individual original features
- C. Are class labels
- D. Are randomly generated variables

41. Reducing a dataset from 50 dimensions to 2 dimensions using PCA is particularly useful for:

- A. Visualization
- B. Increasing dimensionality
- C. Generating labels
- D. Increasing the number of samples

42. What is the main trade-off when reducing dimensions using PCA?

- A. Some information/variance may be lost
- B. The number of samples always decreases
- C. The data becomes categorical

- D. The covariance matrix becomes larger

43. Suppose PC1 is represented as:

$$\mathbf{PC1 = 0.8X_1 + 0.5X_2 - 0.2X_3}$$

What do the coefficients primarily indicate?

- A. Contributions/weights of the original features to PC1
- B. Classification probabilities
- C. Feature variances
- D. Number of samples

44. If a feature has a large absolute loading on PC1, it suggests that:

- A. The feature contributes strongly to PC1
- B. The feature must be removed
- C. The feature is always the most important feature for prediction
- D. The feature has zero variance

45. If two original features have very similar loadings on PC1, this may suggest:

- A. They contribute similarly to that component
- B. They are necessarily identical
- C. They must have zero covariance
- D. PCA has failed

46. PCA components are often difficult to interpret because:

- A. They combine information from multiple original variables
- B. They contain no numerical values
- C. They are always categorical
- D. They contain only class labels

47. Which statement about PCA is most appropriate?

- A. PCA guarantees that the most predictive features are selected
- B. PCA identifies directions of high variance
- C. PCA requires class labels
- D. PCA always improves model accuracy

48. PCA can be sensitive to:

- A. Feature scaling and outliers
- B. Only class labels

- C. The names of the features
- D. The order in which rows are stored

49. If an outlier has extremely large values, it may:

- A. Strongly influence the principal components
- B. Have no effect on PCA
- C. Automatically be removed by PCA
- D. Become a class label

50. Which statement best describes PCA?

- A. PCA selects the original features with the highest variance
- B. PCA creates new orthogonal directions that capture variance in the data
- C. PCA converts continuous data into categorical data
- D. PCA is primarily a classification algorithm